REGC algorithm as implemented in `gcmc_re`

This document states the replica-exchange grand-canonical (REGC) Monte Carlo algorithm that `gcmc_re` implements, with every acceptance rule written out, and relates it to

> Y. Zhou, M. Scheffler, L. M. Ghiringhelli, *Determining surface phase diagrams including anharmonic effects*, Phys. Rev. B **100**, 174106 (2019) — referred to below as **[ZSG]**.

Equations here are numbered (1), (2), …; equations of the paper are cited as [ZSG Eq. n].

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0. Notation

Symbol	Meaning	Code
$T,\ \beta = 1/k_B T$	temperature (energy units, $k_B = 1$)	`ThermoState.temperature`, `.beta`
μ	chemical potential of the exchanged species	`ThermoState.mu`
$\Lambda(T)$	thermal de Broglie wavelength, $h/\sqrt{2\pi m k_B T}$	`ThermoState.lambda_cubed` $= \Lambda^3$
$E(\mathbf{R})$	potential energy of configuration $\mathbf{R}$ (calculator + external field)	`Replica.energy`, `ParticleSystem.total_energy()`
N	number of mobile (exchangeable) particles	`ParticleSystem.n`
V_{box}	region where mobile particles may live	slab: $L_x L_y\,(0, L_z]$
V_{ex}	**exchange zone**: region where insertions are placed and deletions are drawn from	`ParticleSystem.volume`
N_{ex}	number of mobile particles currently inside V_{ex}	`len(exchange_zone_indices())`
(l, m)	grid index: T_l, μ_m	`sim.grid[(it, imu)]`

One $\Lambda^3(T)$ convention, used everywhere. Λ^3 enters three places: the insertion/deletion rules (§2), the replica-exchange rule (§3) and the MBAR reduced potential (§5). All three read the same function, `lambda_cubed_of_T`, passed to `ReplicaExchangeGCMC` (and picked up from `sim` by `MBARSurfaceAnalysis`):

- **Reduced LJ units** (default): $\Lambda^3 \equiv 1$ at every T.
- **Real units**: $\Lambda^3(T) = \left(h^2/2\pi m k_B T\right)^{3/2}$ (`units.thermal_wavelength_cubed_angstrom3`), which must be valid at *any* T, not only at grid temperatures, because MBAR evaluates it off-grid.

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1. Target ensemble

Each grid point (l, m) samples the grand-canonical density [ZSG Eq. 1]

$$\mathcal{N}_{\mu,V,T}(\mathbf{R}; N) \;\propto\; \frac{e^{\beta\mu N}\, V_{\mathrm{box}}^{N}}{\Lambda^{3N}(T)\, N!}\; e^{-\beta E(\mathbf{R};N)} \tag{1}$$

with positions in V_{box} (normalized coordinates). The extended ensemble is the product over all $L \times M$ grid points [ZSG Eq. 4]:

$$Q_{\mathrm{ext}} = \prod_{l=1}^{L}\prod_{m=1}^{M}\sum_{N}\frac{e^{\beta_l \mu_m N}}{\Lambda^{3N}(T_l)\, N!}\int_{V_{\mathrm{box}}^{N}} d\mathbf{R}\; e^{-\beta_l E(\mathbf{R};N)}. \tag{2}$$

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2. Local moves (one grid point, fixed T, μ)

A *sweep* (`Replica.sweep`) performs `sweeps_per_cycle` moves. Each move is drawn independently with relative probabilities `p_displace : p_insert : p_delete : p_displace_substrate` (renormalized). `p_insert = p_delete` is needed for detailed balance and is the default.

2.1 Displacement (`_move_displace`) [ZSG Eq. 7]

Pick a mobile particle i uniformly and propose $\mathbf{r}_i' = \mathbf{r}_i + \boldsymbol{\delta}$, with $\boldsymbol{\delta}$ uniform in $[-\delta_{\max}, \delta_{\max}]^3$:

$$P_{\text{acc}}(\mathbf{r}_i \to \mathbf{r}_i') = \min\left[1,\ e^{-\beta\,\Delta E}\right], \qquad \Delta E = E(\mathbf{r}_i', \mathbf{r}^{N-1}) - E(\mathbf{r}_i, \mathbf{r}^{N-1}). \tag{3}$$

A move outside $(0, L_z]$ has $E = +\infty$ and is always rejected. Particles may move freely in and out of V_{ex}.

`_move_displace_substrate` applies the same rule (3) to a displaceable substrate atom.

2.2 Diffusion step of the REGC loop (`Replica.diffuse`) [ZSG §II A 3]

In the [ZSG] loop (§4), each step ends with canonical (fixed N, V, T) sampling, run in parallel over replicas. Three modes are available:

- **`'mc_moves'`**: n single displacement attempts of rule (3), each of a uniformly chosen movable atom. This is [ZSG]'s LJ protocol ("ten sequential MC moves").
- **`'mc'`**: n MC cycles. One cycle is N_{mov} attempts of rule (3), where N_{mov} is the number of mobile atoms (plus substrate atoms if `displace_substrate=True`), so on average each atom is attempted once per cycle.
- **`'hmc'`**: n Hybrid MC trajectories. Draw momenta $\mathbf{p} \sim$ Maxwell–Boltzmann(T), integrate n_{LF} leapfrog steps of size Δt for all mobile and substrate atoms together, and accept with $\min[1, e^{-\beta \Delta H}]$, $H = E + \sum_j p_j^2/2m_j$. The result is exact for any Δt. This mode is an extension beyond [ZSG].
- **`'md'`**: n steps of NVT molecular dynamics with ASE (`Replica._diffuse_md`), as in the paper's ab initio REGCMD:
  - The thermostat is Bussi stochastic velocity rescaling (default; [ZSG] use $\tau = 20$ fs and a 1 fs time step) or Langevin.
  - Velocities are redrawn from Maxwell–Boltzmann at T_l at the start of each segment. This is exact because $\mathbf{p}$ and $\mathbf{R}$ are independent in the canonical ensemble, and it handles freshly inserted atoms and configurations just swapped in from another T.
  - Frozen atoms are held by `FixAtoms`.
  - The hard walls of V_{box} are enforced by specular reflection after every step ($z \to 2z_{\text{wall}} - z$, $v_z \to -v_z$), the analogue of the reflecting boundaries that [ZSG] impose with PLUMED.
  - After the segment, E is recomputed with the system's calculator.
  - With `LammpsCalculator` (`gcmc_re/lammps_backend.py`) the whole segment runs natively in LAMMPS: `velocity create`, `fix nve` + `fix temp/csvr` (Bussi), `fix wall/reflect`, with frozen atoms excluded. Any LAMMPS pair style works this way (EAM, GRACE-FS `grace/fs`, ACE, ...). It is about $10^3\times$ faster than ASE for an EAM slab.
  - MD has no accept/reject step, so it carries the integrator's time-step bias. The requirements are an `ASECalculatorAdapter` and T given as $k_B T$ in eV.

The mixed sweep scheme `run` (§4.2) does not use `diffuse`. It interleaves the single moves of §2.1 (or HMC, with `hmc_enabled=True`) with GC moves.

2.3 Insertion (`_move_insert`) [ZSG Eq. 3]

Draw $\mathbf{r}_{\text{new}}$ uniformly in V_{ex} (`random_insertion_position`) and accept with

$$P_{\text{acc}}(N \to N+1) = \min\left[1,\ \frac{V_{\text{ex}}}{\Lambda^3(T)\,(N_{\text{ex}}+1)}\ e^{\beta[\mu - (E_{N+1} - E_N)]}\right]. \tag{4}$$

2.4 Deletion (`_move_delete`) [ZSG Eq. 2]

Pick particle i uniformly **among the $N_{\mathbf{ex}}$ particles inside $V_{\mathbf{ex}}$** (no-op if $N_{\mathrm{ex}} = 0$) and accept with

$$P_{\mathrm{acc}}(N \to N-1) = \min\left[1,\ \frac{\Lambda^3(T)\, N_{\mathrm{ex}}}{V_{\mathrm{ex}}}\ e^{-\beta[\mu+(E_{N-1}-E_N)]}\right]. \tag{5}$$

2.5 Why $V_{\mathbf{ex}}$ and $N_{\mathbf{ex}}$ (fix 3)

Treat the particles as an unlabelled set. Then the density (1) is $\pi(\mathbf{R}) \propto e^{\beta\mu N}\Lambda^{-3N}e^{-\beta E}$ per unit volume element (the $N!$ cancels against the labelling). Take a configuration with N_{ex} particles inside V_{ex}:

- the forward move (insertion at $\mathbf{r}_{\mathrm{new}} \in V_{\mathrm{ex}}$) has proposal density $1/V_{\mathrm{ex}}$;
- the reverse move (deleting exactly that particle) has proposal probability $1/(N_{\mathrm{ex}}+1)$, because the candidates are the zone particles *after* insertion.

Detailed balance $\pi_{\mathrm{old}}\, q_{\mathrm{fwd}}\, P_{\mathrm{acc}}^{\mathrm{fwd}} = \pi_{\mathrm{new}}\, q_{\mathrm{rev}}\, P_{\mathrm{acc}}^{\mathrm{rev}}$ then gives

$$\frac{P_{\mathrm{acc}}^{\mathrm{fwd}}}{P_{\mathrm{acc}}^{\mathrm{rev}}} = \frac{\pi_{\mathrm{new}}}{\pi_{\mathrm{old}}}\,\frac{q_{\mathrm{rev}}}{q_{\mathrm{fwd}}} = \frac{e^{\beta\mu}}{\Lambda^3}\,e^{-\beta\Delta E}\,\frac{V_{\mathrm{ex}}}{N_{\mathrm{ex}}+1}, \tag{6}$$

which is exactly (4)/(5). Two consequences:

- The stationary distribution is the full grand-canonical ensemble (1) over **all of $V_{\mathbf{box}}$**, not only V_{ex}. For example, an ideal gas has $\langle N\rangle = e^{\beta\mu}V_{\mathrm{box}}/\Lambda^3$. The recorded N is the total N, and that is the N to use in §3 and §5.
- Using V_{ex} with the *total* N, or deleting particles that sit outside V_{ex}, breaks (6). The previous code did both. Insertion into a sub-region also matches [ZSG], which restricts "insertion (and removal)" to a 48 Å surface zone.

Test: `tests/test_paper_consistency.py::test_deletion_restricted_to_insertion_zone_ideal_gas`.

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3. Replica exchange (`ReplicaExchangeGCMC._attempt_exchange`)

Replica a sits at (β_a, μ_a) with configuration $\mathbf{R}_i$ (N_a, E_a), and replica b at (β_b, μ_b) with $\mathbf{R}_j$ (N_b, E_b). The swap exchanges the *configurations*; the thermodynamic states stay on their grid slots. Detailed balance in the extended ensemble [ZSG Eq. 5]:

$$\mathcal{N}(\beta_a,\mu_a,\mathbf{R}_i)\,\mathcal{N}(\beta_b,\mu_b,\mathbf{R}_j)\,P_{\to} = \mathcal{N}(\beta_a,\mu_a,\mathbf{R}_j)\,\mathcal{N}(\beta_b,\mu_b,\mathbf{R}_i)\,P_{\leftarrow}. \tag{7}$$

Insert (1). The factors V^N and $N!$ appear once on each side and cancel. What remains is

$$P_{\mathrm{acc}}^{\mathrm{swap}} = \min\left[1,\ \left(\frac{\Lambda^3(T_a)}{\Lambda^3(T_b)}\right)^{N_a-N_b}\exp\Big\{(\beta_a-\beta_b)(E_a-E_b)\ -\ (\beta_a\mu_a-\beta_b\mu_b)(N_a-N_b)\Big\}\right]. \tag{8}$$

3.1 Relation to [ZSG Eq. 6]

- **Λ prefactor (fix 1).** For a physical $\Lambda \propto \beta^{1/2}$, $(\Lambda_a^3/\Lambda_b^3)^{N_a-N_b} = (\beta_a/\beta_b)^{\frac{3}{2}(N_a-N_b)}$, which is the paper's factor. The code computes the prefactor from `state.lambda_cubed`, the same value used in (4)/(5), rather than hard-coding $(\beta_a/\beta_b)^{3/2}$. In reduced units ($\Lambda^3 \equiv 1$) the prefactor is 1. The previous code always applied the β factor, which was inconsistent with the $\Lambda^3 = 1$ insert/delete moves.

- **Sign of the μ term.** As printed, [ZSG Eq. 6] has $+(\beta_l\mu_m - \beta_k\mu_o)(N_{l,m} - N_{k,o})$. Working out (7) from (1) gives the minus sign in (8). Sanity check: at equal β with $\mu_a > \mu_b$ and $N_a > N_b$, moving the larger configuration to the lower μ must be *penalized*, and only the minus sign does that. The code uses (8).

Tests: `test_exchange_acceptance_uses_same_lambda_as_gcmc` (rule (8) exactly, both Λ conventions) and `test_replica_exchange_ideal_gas_density` (end-to-end ideal-gas check).

3.2 Swap patterns ([ZSG] Fig. 2, `ReplicaExchangeGCMC.swap_pairs`)

Each pattern is a set of **disjoint** pairs, so all of its swaps can be attempted at once. Each attempt uses (8). With offset $o \in \{0, 1\}$:

Type	Pairs	Fig. 2
`'T'`	$(l,m) \leftrightarrow (l+1,m)$, $l = o, o+2, \ldots$; all m	(a)
`'mu'`	$(l,m) \leftrightarrow (l,m+1)$, $m = o, o+2, \ldots$; all l	(b)
`'cross_left'`	$(l,m+1) \leftrightarrow (l+1,m)$, $l = o, o+2, \ldots$; all m	(c)
`'cross_right'`	$(l,m) \leftrightarrow (l+1,m+1)$, $l = o, o+2, \ldots$; all m	(d)

Over all types and both offsets, every replica is connected to all of its 3–8 neighbours ([ZSG]: "between three and eight neighbors"). Fig. 2 draws only offset 0 (pairs T_1T_2, T_3T_4). With that offset alone the grid would split into disconnected $2 \times M$ blocks, since T_2 would never swap with T_3. The code therefore draws the offset at random as well.

Detailed balance. The choice of pattern and offset does not depend on the current configurations. The pairs in a pattern are disjoint, and each pair's swap satisfies (7). So every RE move satisfies detailed balance in the extended ensemble (2).

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4. Simulation loop

4.1 [ZSG] Fig. 1 scheme: `ReplicaExchangeGCMC.run_regc`

```
repeat for step = 1, 2, … , n_steps:
    draw rand ~ U[0,1)                          (one draw, shared by all replicas)
    if rand < x0:                               particle insertion/removal
        every replica: one GC attempt           removal if y < 1/2, else insertion (Eqs. 4-5)
    else:                                       replica exchange
        draw type ∈ {T, mu, cross_left, cross_right} and offset ∈ {0,1}, uniformly
        attempt every pair of that pattern      (Eq. 8)
    every replica, in parallel: diffusion at fixed N,V,T_l  (MC / HMC / ASE-MD, §2.2)
    if step > burn_in and step % record_every == 0:
        record (E, N) for every replica         (after diffusion, before next GC/RE)
    if step % convergence_every == 0 and ρ(u) unchanged (Eq. 15): stop
```

Execution in blocks. Between two RE moves the replicas are independent. `run_regc` therefore draws the GC/RE branches in order and hands each replica a whole block of steps in one call (`Replica.regc_local_block`), with samples recorded inside the block. A block ends before the next RE move or after a step that needs the main process (trajectory frame, progress line, convergence check). The random draws and their order are exactly those of a step-by-step loop, so results are bit-identical (tested). With $x_0 = 0.99$ a block is about 100 steps long, which makes process-pool parallelism worthwhile.

Which branch x_0 selects. The paper is internally inconsistent here:

- Fig. 1 labels `rand` $\leq$ `x0` $\rightarrow$ *replica exchange* and *no* $\rightarrow$ *particle insertion/removal*.
- The text of §II A says "the system has a probability x_0 to attempt exchanging a particle with the reservoir and probability $(1 - x_0)$ to perform a RE move".
- The parameters agree with the **text**. For the LJ surface, $x_0 = 0.99$ together with "configuration swaps were attempted every 100 REGC steps" $(1 - x_0 = 0.01)$. For the Si clusters $x_0 = 0.9$.

`run_regc` follows the text: $x_0 = P(\textbf{GC attempt})$, default 0.99. To use Fig. 1's convention, pass `x0 = 1 - x0_fig`.

Convergence ([ZSG] §III A: "no change in the density of reduced-energy states $\rho(U)$ with increasing simulation steps"). Every `convergence_every` steps, for each grid point k, the reduced energies u of Eq. (9) of all samples recorded so far are histogrammed on common bins. The same is done for the samples that existed at the previous check. The run stops when

$$\max_k \ \tfrac{1}{2} \sum_{\text{bins } b} |\rho_k^{\text{now}}(b) - \rho_k^{\text{prev}}(b)| \ < \ \texttt{convergence_tol}, \tag{15}$$

that is, when the total-variation change of every $\rho_k(u)$ is small. `n_steps` is a cap. `run_regc` returns whether the criterion was met, and `convergence_history` stores the values. `swap_acceptance()` reports the acceptance per pattern. [ZSG] tune the (T, μ) spacing to reach about 25%.

4.2 Mixed sweep scheme: `ReplicaExchangeGCMC.run`

```
for cycle = 1 … n_cycles:
    every replica, in parallel: sweeps_per_cycle moves drawn from
        {displace|HMC, insert, delete, displace_substrate}
    if cycle % exchange_every == 0:
        for type in (T, mu, cross_left, cross_right):   # diagonals: diagonal_swaps=True
            for offset in (0, 1): attempt all pairs of that pattern
    record (E, N) every record_every cycles after burn-in
```

This scheme mixes GC moves into the sweep and visits the swap patterns in a fixed order. Each move satisfies detailed balance, and a fixed-order composition of such moves still leaves (2) invariant (global balance). Both loops therefore sample the same ensemble. §4.1 reproduces the paper's protocol; §4.2 is usually more efficient for cheap force fields.

The YAML runner selects between them with `replica_exchange.scheme: sweep | regc`. The `regc` scheme reads `n_steps`, `x0`, `diffusion`, `diffusion_steps`, `record_every`, `burn_in_steps`, `convergence_every`, `convergence_tol`, and an optional `md:` block (`timestep_fs`, `thermostat`, `taut_fs`, `friction_per_fs`).

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5. MBAR reweighting (`analysis.MBARSurfaceAnalysis`) [ZSG §II B]

5.1 Reduced potential (fix 2)

MBAR needs, for every sample n and every state $k = (T, \mu)$, the reduced potential $u_k(n) = -\ln q_k(n)$ up to a sample-independent constant that may depend on k. From (1), the factors $V^N/N!$ are the same at every state, so they cancel between states, but $\Lambda^{-3N}(T)$ does not. Hence

$$u(\mathbf{R}; \mu, \beta) = \beta \left[E(\mathbf{R}) - \mu N(\mathbf{R})\right] \ + \ N(\mathbf{R}) \ \ln \Lambda^3(T). \tag{9}$$

[ZSG Eq. 9] omits the last term. This is consistent only if Λ is absorbed into μ or is T-independent. Otherwise it contradicts the Λ factor in their own Eq. 6. The code now uses (9) both to build `u_kn` and for off-grid states (`_u_of_state`), evaluating Λ^3 through `lambda_cubed_of_T` at arbitrary T. In reduced units the extra term vanishes.

Test: `test_mbar_includes_lambda_cubed_term` (exact Poisson ideal-gas samples with a T-dependent Λ, reweighted to an unsampled (T, μ)).

5.2 Estimators

With $q(\mathbf{R}; \mu, \beta) = e^{-u}$, MBAR solves self-consistently for the normalizations $\hat{c}_{l,m}$ [ZSG Eq. 10]:

$$\hat{c}_{l,m} = \sum_{n=1}^{N_{\mathrm{tot}}} \frac{q(\mathbf{R}_n; \mu_m, \beta_l)}{\sum_{l',m'} \Lambda_{l',m'}\, \hat{c}_{l',m'}^{-1}\, q(\mathbf{R}_n; \mu_{m'}, \beta_{l'})}, \tag{10}$$

where $\Lambda_{l',m'}$ is the **number of samples** recorded at state (l', m') (the paper's notation; not the wavelength). An observable A at any (μ, β) is then

$$\langle A \rangle_{\mu,\beta} = \frac{\sum_n A(\mathbf{R}_n)\, q(\mathbf{R}_n; \mu, \beta) \,/\, D_n}{\sum_n q(\mathbf{R}_n; \mu, \beta) \,/\, D_n}, \qquad D_n = \sum_{l',m'} \Lambda_{l',m'}\, \hat{c}_{l',m'}^{-1}\, q(\mathbf{R}_n; \mu_{m'}, \beta_{l'}). \tag{11}$$

`mean_N` uses $A = N$. For phases defined by an indicator χ_i (e.g. $N = N_i$), the probability is $p_i = \langle \chi_i \rangle$ and the phase free energy is [ZSG Eq. 8]

$$f_i(\mu, T) = -k_B T \, \ln p_i(\mu, T), \tag{12}$$

with the stable phase at (μ, T) being $\arg\min_i f_i = \arg\max_i p_i$.

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6. From μ to pressure

For a monatomic ideal-gas reservoir, [ZSG §III A] gives

$$\beta\mu = \ln\!\big(\beta p\, \Lambda^3\big) \quad \Longleftrightarrow \quad \mu = k_B T \ln \frac{p}{p_0(T)}, \qquad p_0(T) = \frac{k_B T}{\Lambda^3(T)}. \tag{13}$$

This uses the **same** $\Lambda^3(T)$ as §2–§5. `pressure.p0_ideal_gas_wavelength` implements it. `pressure.p0_reduced_units` ($p_0 = T$) is the $\Lambda^3 \equiv 1$ case.

For an atom exchanged with a diatomic reservoir (e.g. H from H_2), `reservoir.py` uses the standard atomistic-thermodynamics form (cf. [ZSG Eq. 15]):

$$\mu_{\mathrm{H}}(T, p) = \tfrac{1}{2}\left[\mu_{\mathrm{H_2}}(T, p) - D_0\right] + e_{\mathrm{atom}}, \qquad \mu_{\mathrm{H_2}} = k_B T \ln \frac{p\, \Lambda_{\mathrm{H_2}}^3}{k_B T} - k_B T \ln(q_{\mathrm{rot}}\, q_{\mathrm{vib}})\,, \tag{14}$$

with D_0 and e_{atom} computed with the same calculator that evaluates the adsorbed atoms. e_{atom} is the calculator's energy of one isolated atom. All GCMC energy differences are measured on the calculator's scale, so an atom far from the surface costs $E_{N+1} - E_N = e_{\mathrm{atom}}$ in (4). Equivalently, μ_{H} is half the H_2 chemical potential measured from the calculator's own $E_{\mathrm{H_2}}^{\mathrm{min}} + \mathrm{ZPE}$. This is the role E_{DFT} plays in [ZSG Eq. 15]. For EMT's H, $e_{\mathrm{atom}} = 3.21$ eV, while the native `LJCalculator` has $e_{\mathrm{atom}} = 0$. Omitting it shifts every μ_{H} by $-e_{\mathrm{atom}}$. In (4), (5), (8) and (9), the Λ is that of the **exchanged species** (the H atom), since that is the particle whose number fluctuates.

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7. The LJ surface of [ZSG] §III A (`examples/12_paper_lj_surface.py`)

Item	[ZSG]	Here
Surface	frozen A_{18}, 2-layer fcc(111), hexagonal 3×3 cell, lattice vectors 11.489 Å long	`SlabSystem(cell_xy=...)`, exact minimum image for non-orthogonal cells
Interactions	LJ 12-6: $\epsilon_{AB} = 0.66$ eV, $\sigma_{AB} = 1.91$ Å; $\epsilon_{BB} = 0.01$ eV, $\sigma_{BB} = 1.2$ Å; no A–A	`PairLJCalculator` (sees frozen atoms); cutoff 2.5σ, shifted (not stated in [ZSG])
System volume	the 48 Å "surface zone" above the topmost A atoms, all x, y	box = zone: top A layer at $z = 0$ (hard wall), $L_z = 48$ Å; B confined to it
Grid	$T = 200 \ldots 650$ K (step 50), $\mu = -2.4 \cdots - 0.9$ eV (step 0.1): 160 replicas	same
Protocol	$x_0 = 0.99$; 10 MC moves per step; 1.2×10^5 steps; sample every 100 steps	`run_regc(x0=0.99, diffusion='mc_moves', diffusion_steps=10, record_every=100)`
Mass of B	not stated	$m_B = 16.48$ amu, from the Fig. 3 caption: (200 K, −0.9 eV) $\equiv 2.03 \times 10^{-17}$ atm via Eq. (13)
Displacement amplitude	not stated	0.5 Å

Confining B to the zone matters. If B may leave it (only insertion restricted, §2.5), B ($\sigma_{AB} = 1.91$ Å) slips through the hollows of the top layer (A–A distance 3.83 Å). Most of it then ends up absorbed between the two A layers, which roughly doubles the coverage and erases the paper's layer plateaus. With B confined, single replicas give the following:

- (200 K, −0.9 eV): 18 B in the top-layer hollows ($z \approx 0$), about 27 in a second layer ($z \approx 0.75$–1.25 Å), and a few above, $N_B \approx 61$. [ZSG] give 59.
- (650 K, −2.4 eV): pristine.

The script compares two phase diagrams, both labelling phases by N_B:

- **MBAR@REGC:** `MBARSurfaceAnalysis.most_probable_N`, i.e. $\arg\max_N P(N)$ from Eq. (12).
- **aiAT@REGC:** the lowest sampled energy per N_B in [ZSG Eq. 13], with vibrations neglected.

It reports how these diagrams relate to the paper's results:

- RE acceptance; [ZSG] report about 25%.
- Whether the paper's phases $N_B = 0, 18, 45, 59, 85$ appear.
- Agreement between the two diagrams at $N_B \leq 18$; [ZSG] say they "almost coincide".

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7. Summary of changes relative to [ZSG]

#	Item	Status
1	Swap prefactor hard-coded as $(\beta_a/\beta_b)^{3/2\,\Delta N}$, wrong when $\Lambda^3 \equiv 1$	**Fixed:** Eq. (8) with `lambda_cubed` (`engine.py`)
2	MBAR reduced potential missing $N \ln \Lambda^3(T)$	**Fixed:** Eq. (9) (`analysis.py`); `lambda_cubed_of_T` in runner and examples 05–07 made valid at any T
3	Deletion drew from all particles and used total N with restricted V_{ex}	**Fixed:** Eqs. (4)–(5) with N_{ex} (`systems.exchange_zone_indices`, `replica.py`)
—	Sign of μ term in [ZSG Eq. 6]	Paper typo; code follows the derivation (8)
4	Diagonal swaps; one random swap type per RE move ([ZSG] Fig. 2)	**Implemented:** `swap_pairs`, `_exchange_round` (§3.2); random offset added for connectivity
5	Fig. 1 loop: x_0 GC vs RE → diffusion → sample → convergence	**Implemented:** `run_regc` (§4.1); $x_0 = P(\text{GC})$ as in the text (Fig. 1's labels are swapped)
6	Canonical diffusion via thermostatted MD	**Implemented:** `diffuse('md')` with ASE Bussi/Langevin NVT (§2.2)
7	Convergence by unchanged $\rho(U)$	**Implemented:** Eq. (15)
—	Mixed sweep scheme `run`	Kept as an alternative (§4.2)
8	Reservoir μ_{H} missing the calculator's isolated-atom energy	**Fixed:** Eq. (14), `e_atom_eV` in `reservoir.py` and the runner